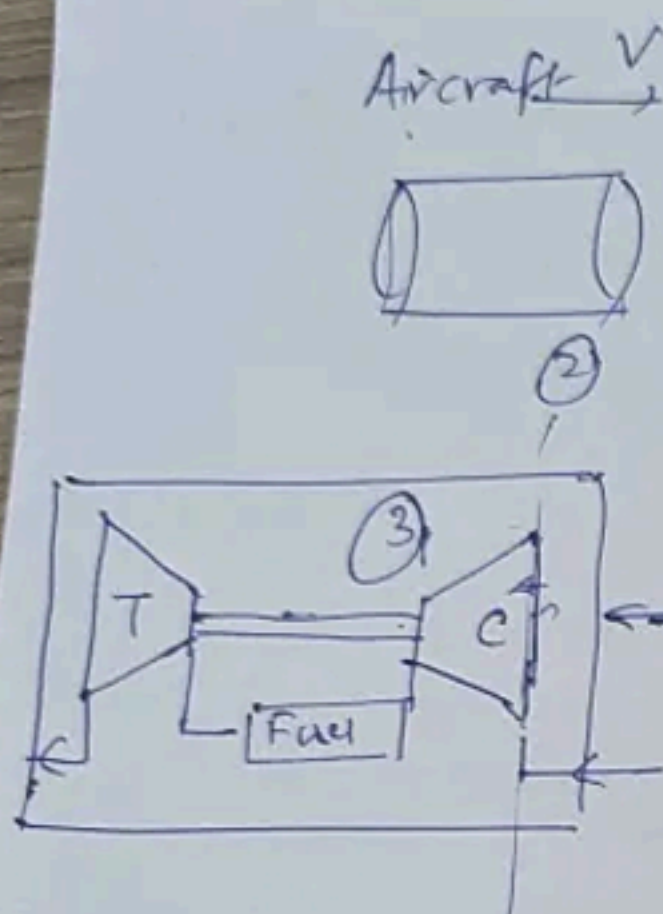




$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \quad q=0$$

$$V_2 \approx 0 \quad (V_2 \ll V_1) \quad w=0$$

$$\rho_1 = \rho_2$$

$$V = 350 \text{ mph} = 880 \text{ km/h} = 245 \text{ m/s}$$

$$\Rightarrow h_1 + \frac{V_1^2}{2} = h_2$$

$$\Rightarrow h_2 - h_1 = \frac{V_1^2}{2}$$

Ideal gas: $c_p(T_2 - T_1) = \frac{V_1^2}{2} \Rightarrow \boxed{T_2 = T_1 + \frac{V_1^2}{2c_p}}$

Ex: $V = 245 \text{ m/s}$ $T = -30^\circ\text{C} = 243 \text{ K}$

$$T_2 = 243 + \frac{(245)^2}{2(1005)} = 243 + 30 = 273 \text{ K}$$

Isentropic $p_1 = 0.28 \text{ bar}$

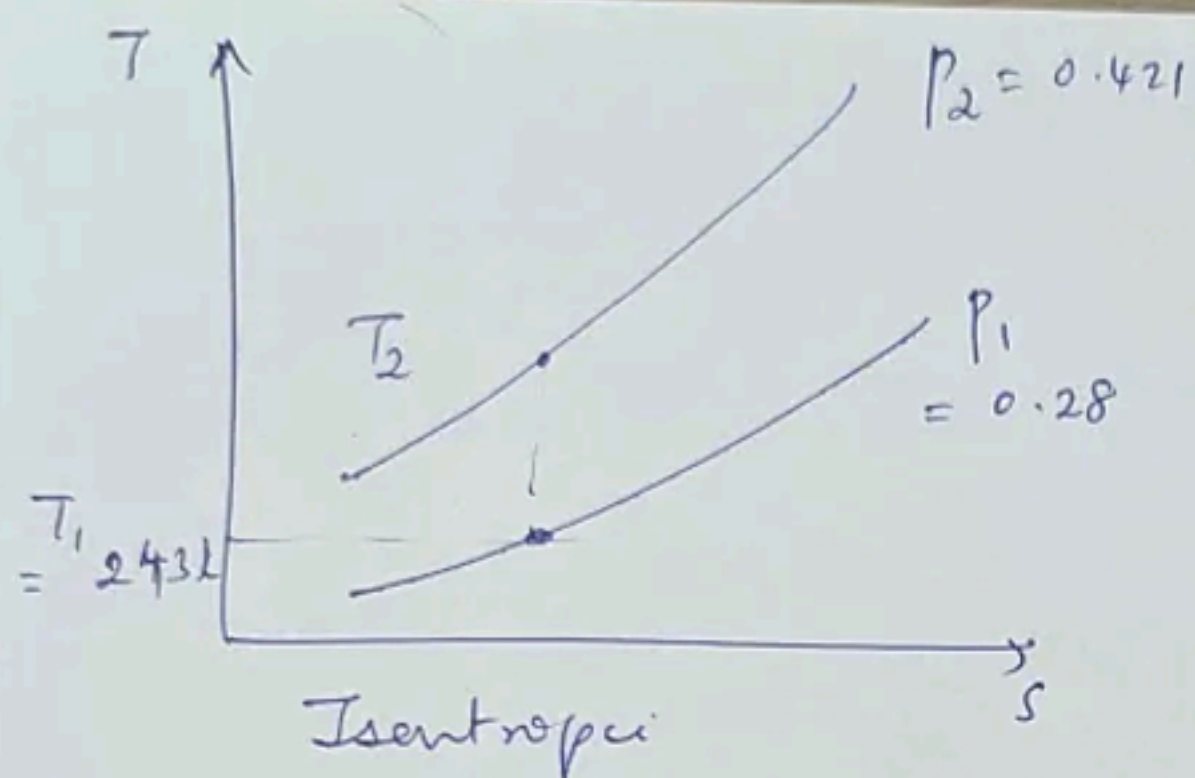
$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{\frac{\gamma}{\gamma-1}} \Rightarrow \frac{p_2}{0.28} = \left(\frac{273}{243}\right)^{\frac{1.4}{0.4}} \Rightarrow \boxed{p_2 = 0.421 \text{ bar}}$$

Density of air at 2

$$\rho_2 = \frac{p_2}{RT_2}$$

$$= \frac{0.421 \times 10^5}{(0.287)(273)}$$

$$= 0.537 \text{ kg/m}^3$$

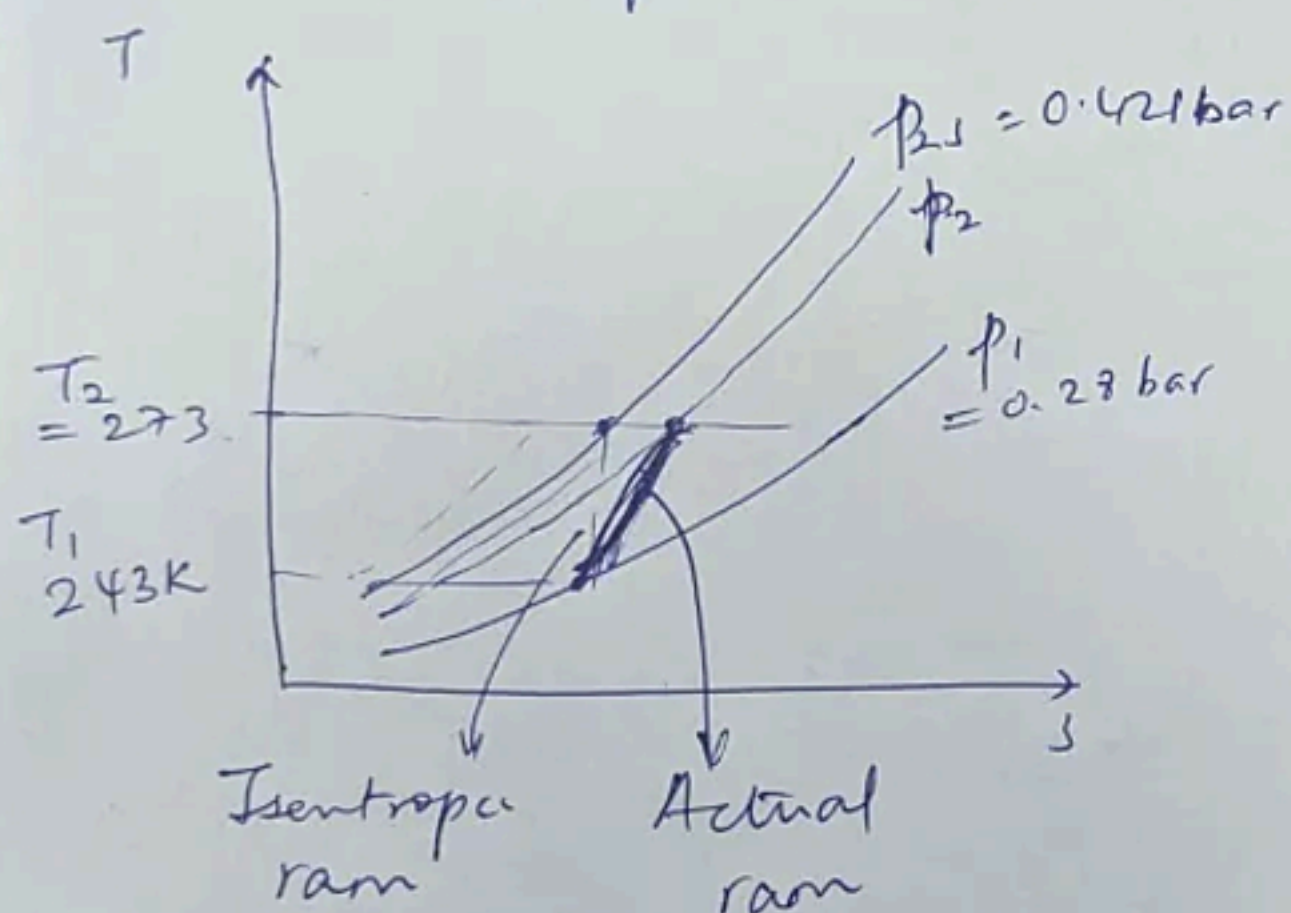


Ex: $\eta_{\text{isentropic}} = 90\%$ (2)
 $p_2 = ?$

$$\eta_{\text{isent}} = 0.9 = \frac{p_2 - 0.28}{0.421 - 0.28}$$

$$\Rightarrow \boxed{p_2 = 0.407 \text{ bar}}$$

$$\rho_2 = \frac{p_2}{RT_2} = \frac{0.407 \times 10^2}{(0.287)(273)} = 0.52 \text{ kg/m}^3$$



$$\boxed{\eta_{\text{isentropic}} = \frac{p_{2s} - p_1}{p_2 - p_1}}$$

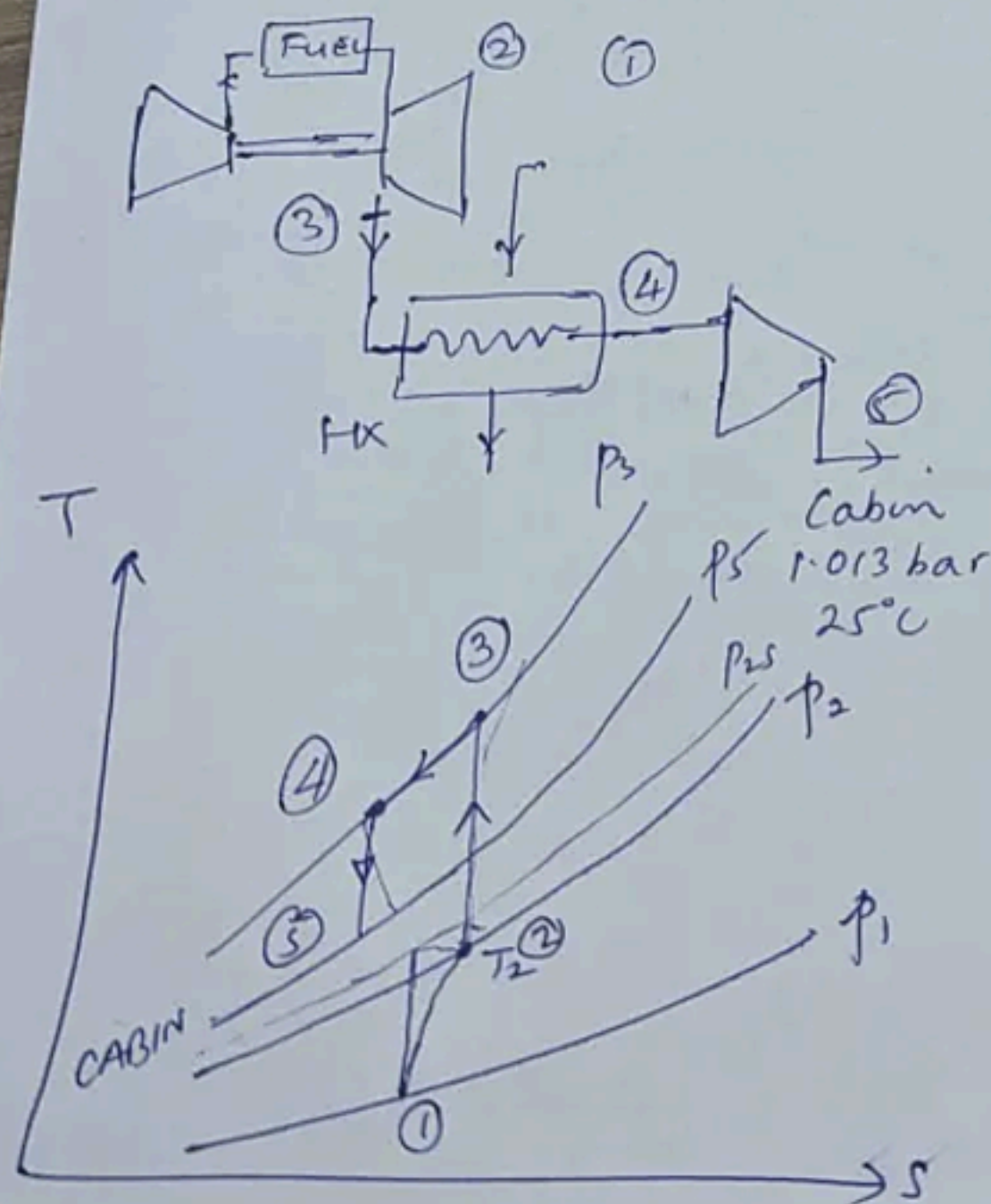
Let compression ratio of main compressor be 8. (3)

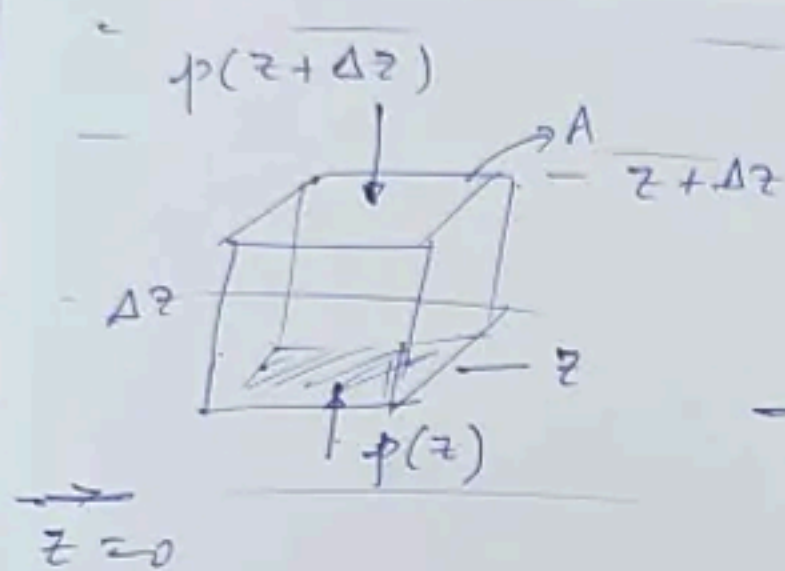
$$\frac{T_3}{T_2} = \left(\frac{P_3}{P_2}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow \frac{T_3}{273} = (8)^{\frac{0.4}{1.4}} \Rightarrow T_3 = 494.5 K$$

$$\frac{P_3}{P_2} = 8 \Rightarrow P_3 = (0.407)^8 = 3.26 \text{ bar}$$

$$\text{Let } T_4 = 350 K$$

$$\frac{T_4}{T_5} = \left(\frac{P_4}{P_5}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow \frac{350}{T_5} = \left(\frac{3.26}{1.013}\right)^{\frac{0.4}{1.4}} \Rightarrow T_5 = 250.7 K$$





$$\begin{aligned} \Sigma F_z &= 0 \quad (4) \\ p(z)A - p(z + \Delta z)A - \rho A \Delta z g &= 0 \\ p(z + \Delta z) - p(z) &= -\rho \Delta z g \\ \Rightarrow \lim_{\Delta z \rightarrow 0} \frac{p(z + \Delta z) - p(z)}{\Delta z} &= -\rho g \\ \Rightarrow \boxed{\frac{dp}{dz} = -\rho g} \end{aligned}$$